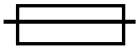
11 PLUS PREPARATION

NON-VERBAL REASONING PAPER

Name:	
Date of Birth:	
Today's Date:	

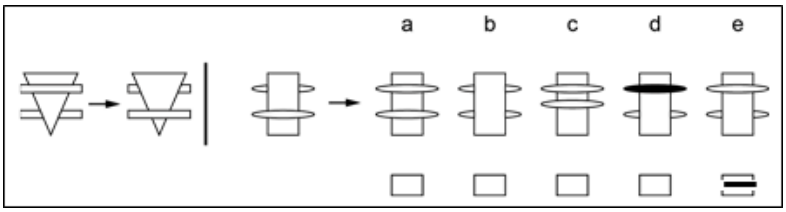
READ THESE INSTRUCTIONS CAREFULLY BEFORE STARTING.

- You have 15 minutes to complete the paper.
- Mark your answer like this: 
- This paper features 13 different question types. An example is given at the start of each question type.
- Each question is worth one mark. If you can't do a question, then move on.
- If you finish early check your answers for mistakes.
- Once the test has started you may not ask for help.

1. LIKE SHAPES / TYPE 1

Work out which of the answer shapes is related to the question shape in the same way that the example shapes are related.

Example



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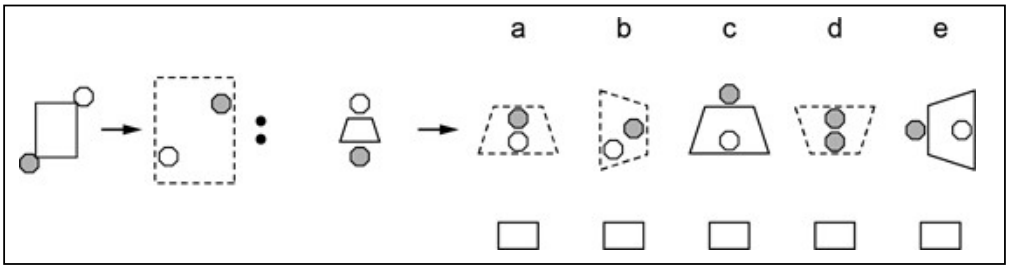

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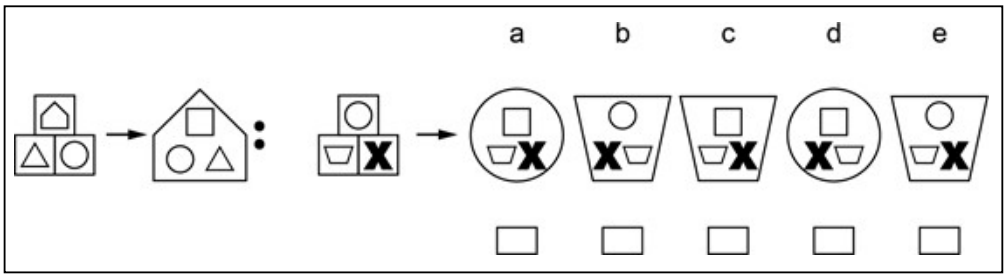

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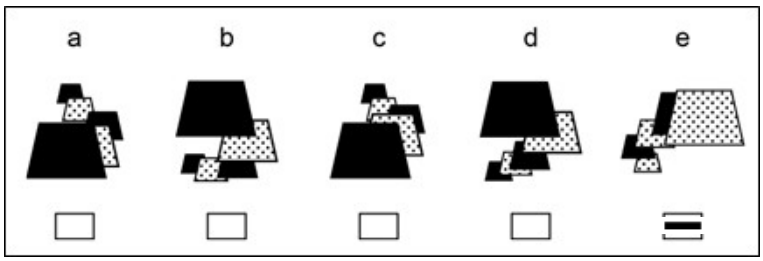
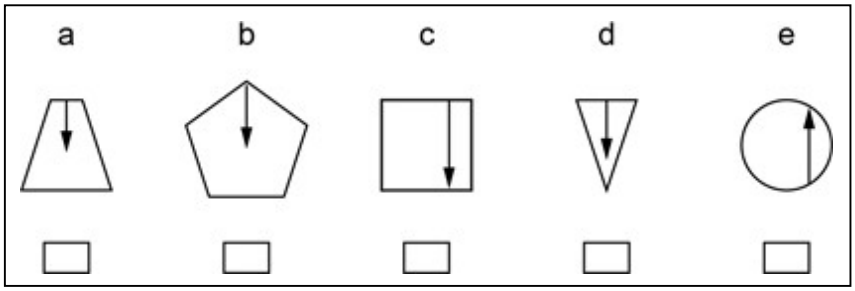
2. ODD ONE OUT / TYPE 2

Work out which of the shapes is different to the rest.

Example



1



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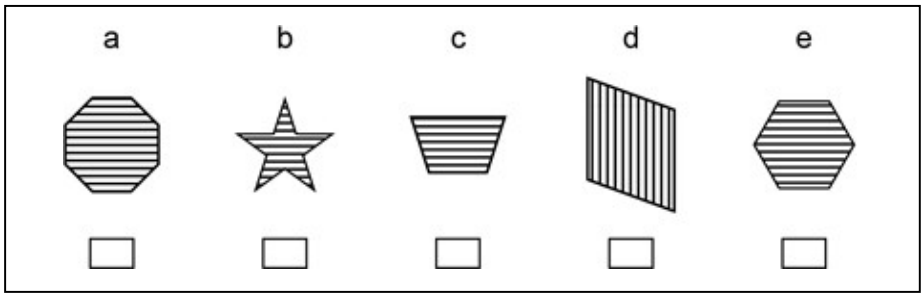
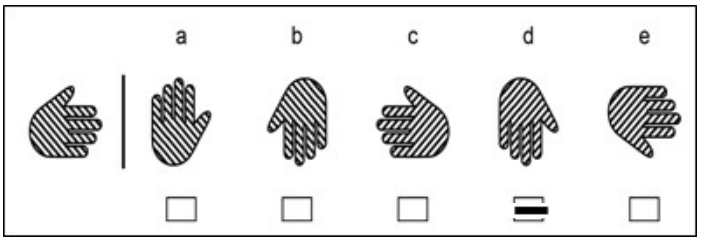


3. ROTATIONS / TYPE 3

Work out which of the answers is a rotation of the question shape.

Example



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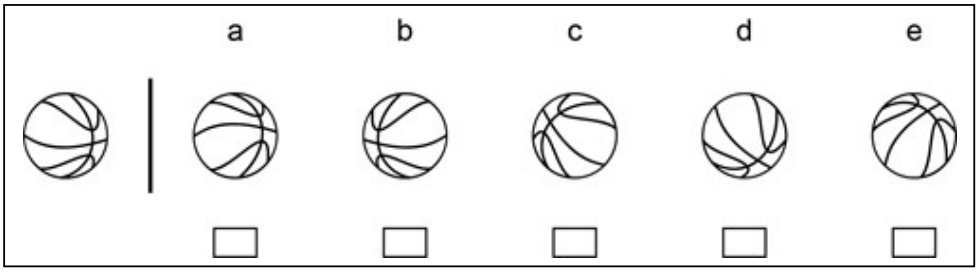

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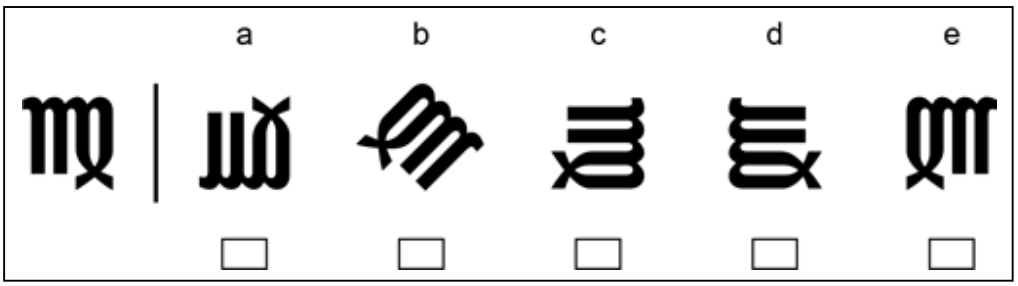
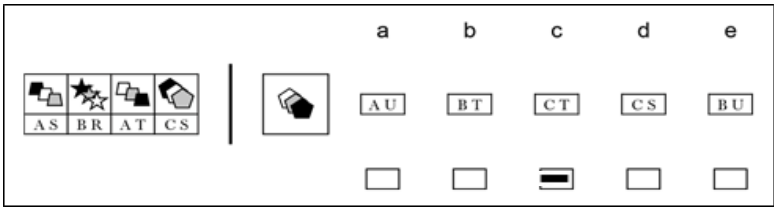


4. CODE BREAKER / TYPE 4

Look at the sequence to work out how the codes match the shapes, then select the correct answer code to match the question shape.

Example



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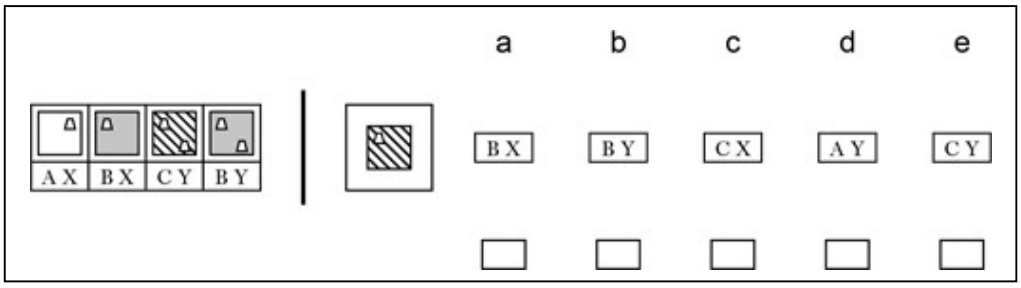

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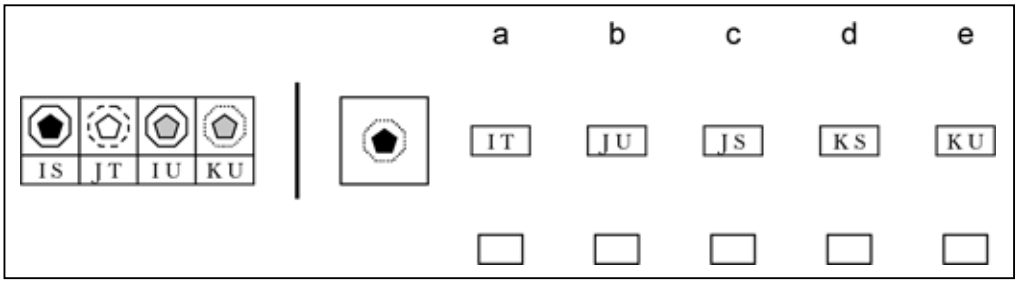
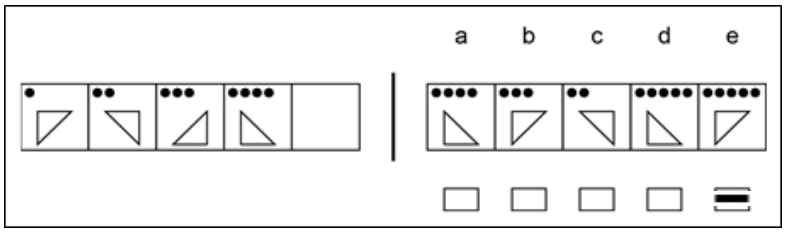


5. MISSING SEQUENCE / TYPE 5

Select the answer that correctly completes the sequence.

Example



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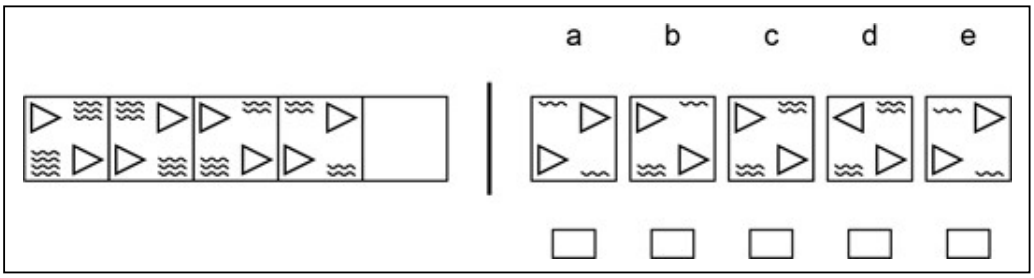

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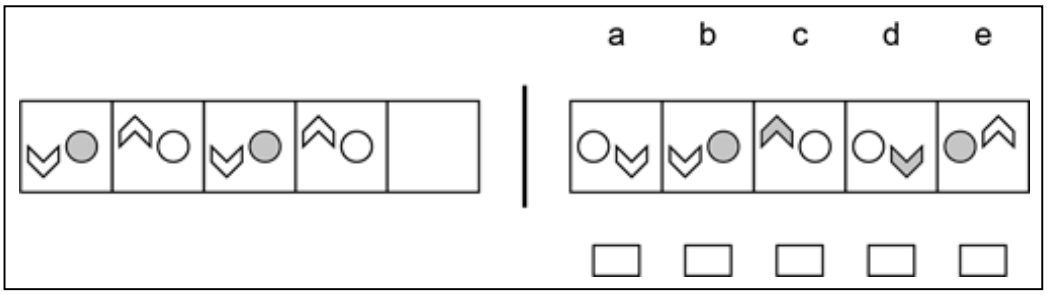

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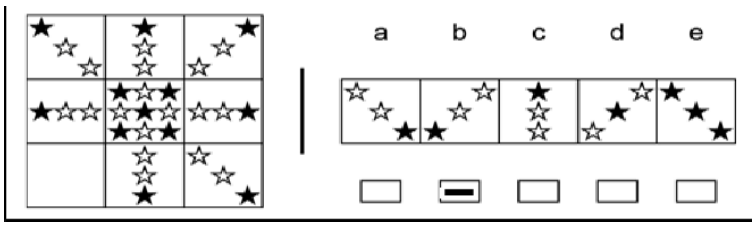
6. MISSING SQUARE / TYPE 6

Select the answer that correctly completes the grid.

Example



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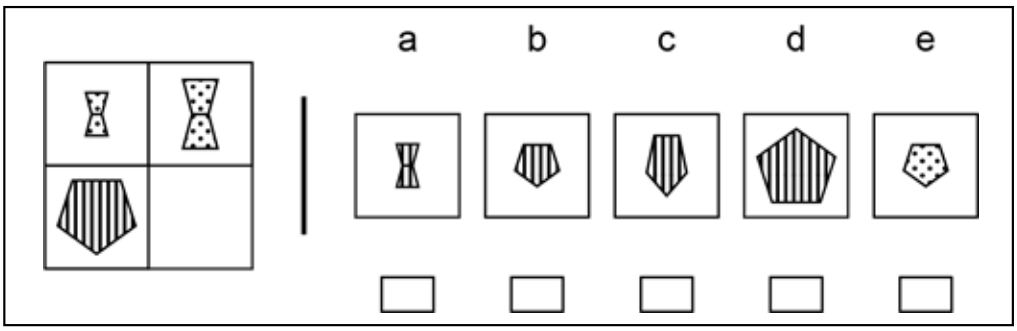

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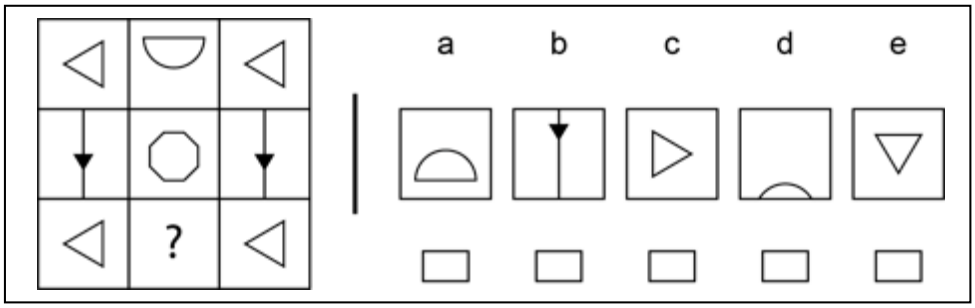
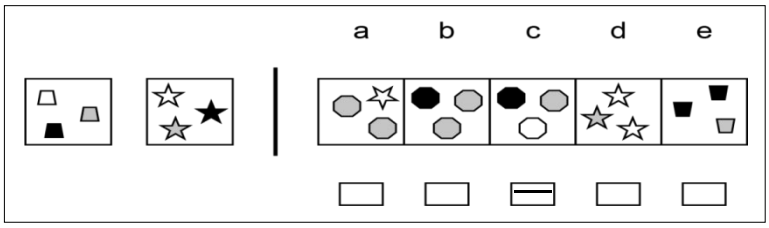


7. MOST LIKE / TYPE 7

Select the answer that is most similar to the two given shapes.

Example



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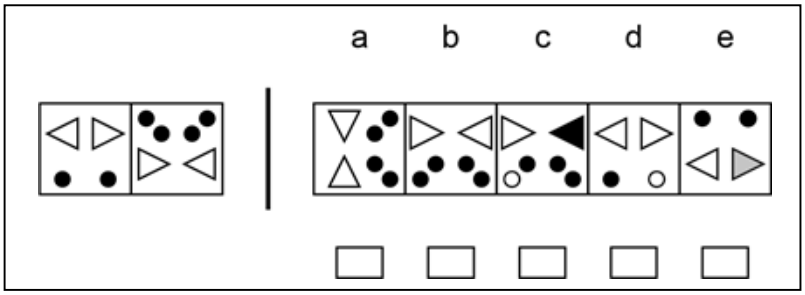

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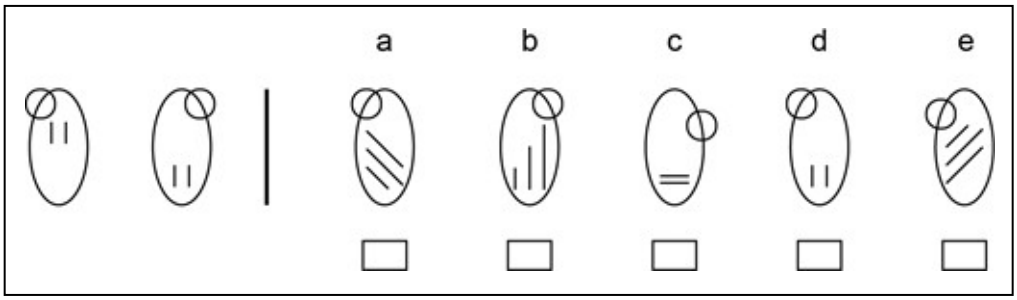

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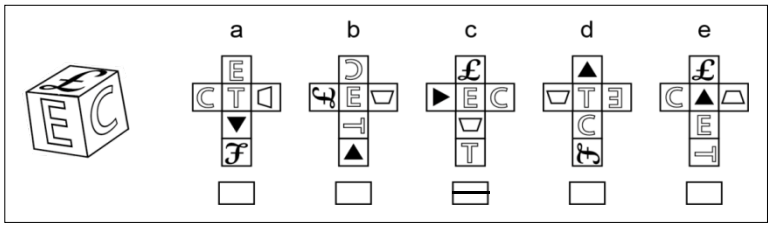
8. NETS / TYPE 8

Select the cube that is made from the given net, or net that makes the given cube.

Example



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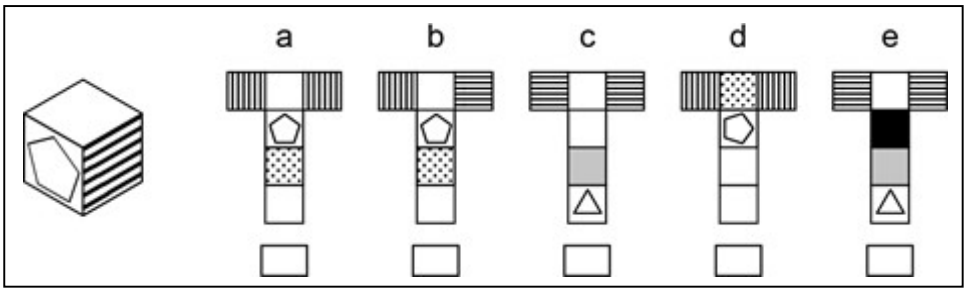

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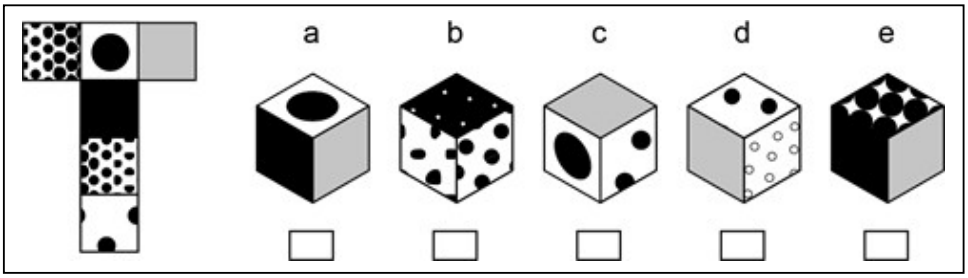
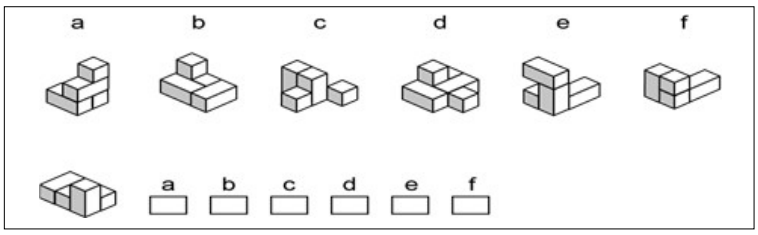


9. 3D SHAPES / TYPE 9

You will need to do one of the following:
Select the answer that is a rotation of the given 3D shape; or
Select the building blocks that make the given 3D shape; or
Select the correct 2D view of the given 3D shape.

Example



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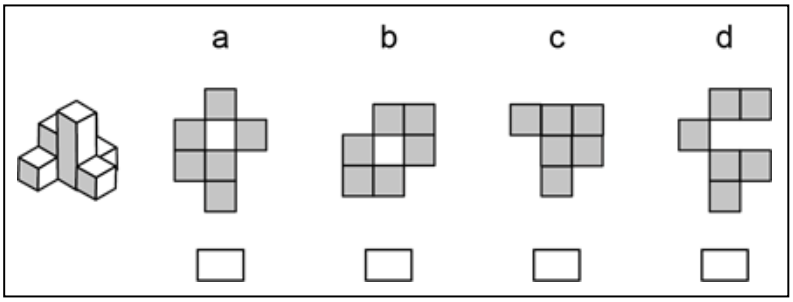

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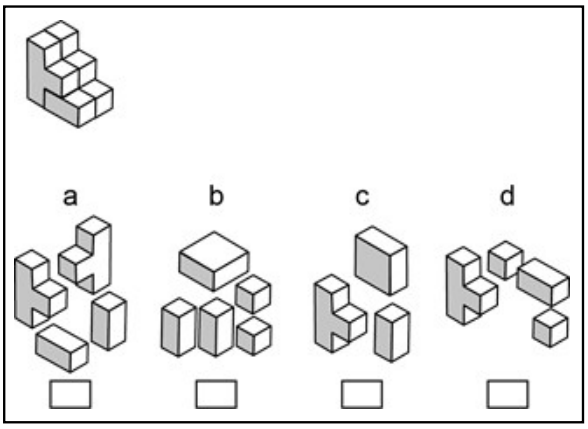

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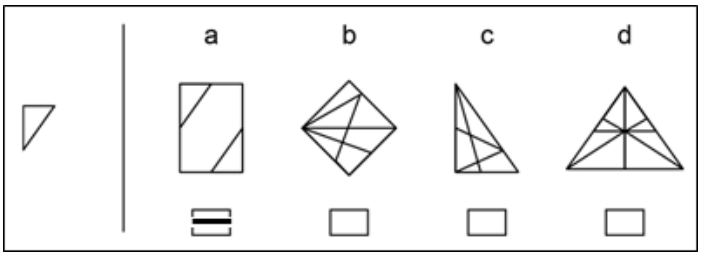
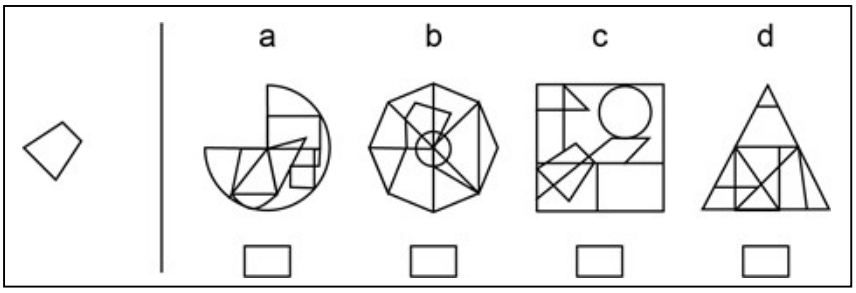
10. HIDDEN SHAPE / TYPE 10

Find the answer option that contains the shape hidden in exactly the same size and orientation.

Example



1



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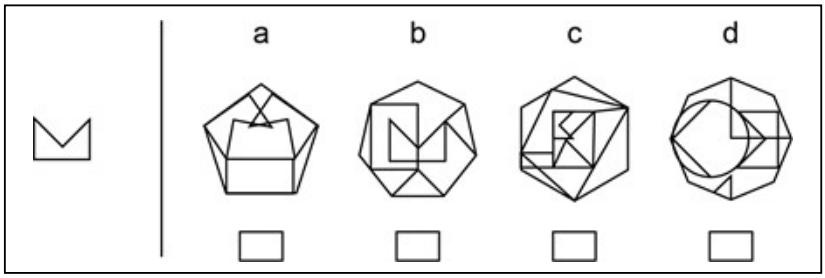
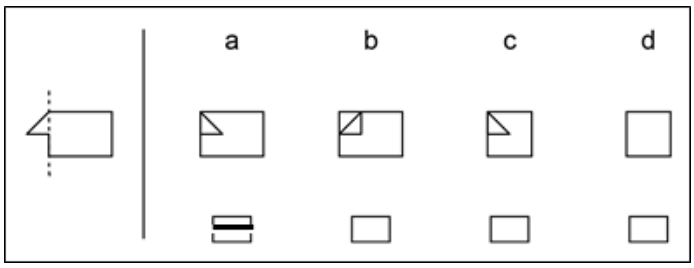


11. FOLD ALONG THE DOTTED LINE / TYPE 11

Select the option that will be formed when the shape is folded along the dotted line.

Example



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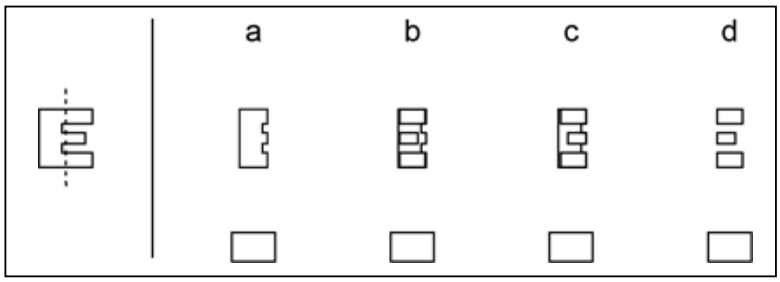
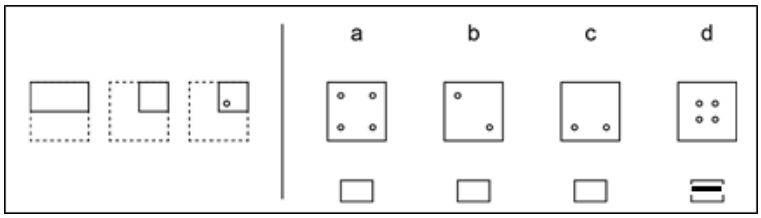


12. HOLE PUNCH / TYPE 12

The sequence shows a shape being folded and then hole punched. Select the option that will be created when the hole punched shape is unfolded.

Example



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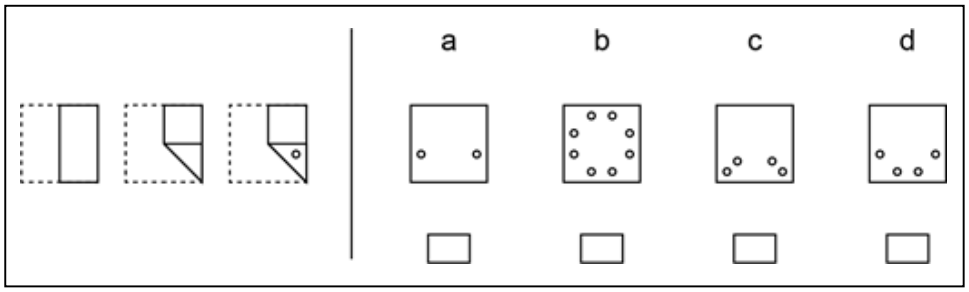
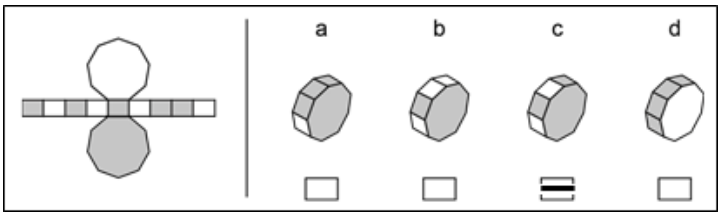


13. SHADED NETS / TYPE 13

Select the 3D shape that is created from the shaded net.

Example



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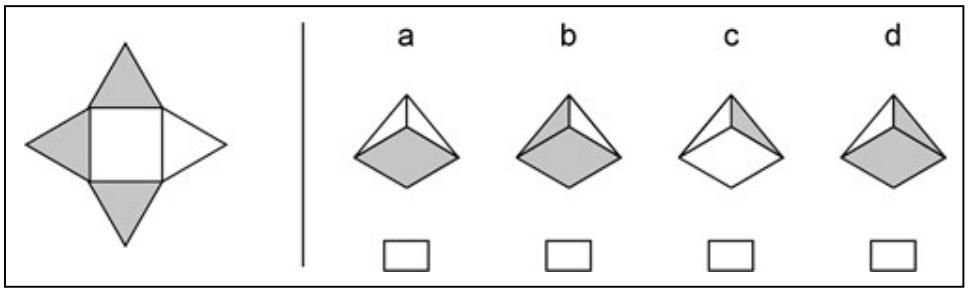


Non-Verbal Reasoning

Like Shapes

1. **A** - (The non-octagon shape becomes bigger and the outline changes from solid to dashed while the octagons then move inside the shape and the two octagons swap colours.)
2. **D** - (The outer shape, the square, changes to be in an inner shape at the top of the 3 inner shapes. The top inner shape in the first set of shapes changes to be the outer shape in the second set. Finally, the bottom two inner shapes swap sides.)

Odd One Out

1. **E** - (The arrows are all pointing downwards except for in E where the arrow is pointing upwards.)
2. **D** - (The lines in D are vertical whereas the lines in all the other shapes are horizontal.)

Rotations

1. **D** - (D is a rotation of around 45 degrees clockwise of the original.)
2. **C** - (C is a rotation of 90 degrees clockwise of the original.)

Code Breaker

1. **C** - (You can work out that B stands for the squares being lightly shaded as there are two of them. Therefore, A stands for the square being white and C stands for the square having diagonal lines. There are two pairs of second letters, X and Y. The only similarity between the two Xs is that there is one trapezium and so the similarity between the two Ys is that there are two trapeziums. So, the final answer is CX.)
2. **D** - (You can work out that I is the solid octagon around the pentagon as there are two of them. Therefore this means that J is the dashed octagon and K is the dotted octagon. There are two U's in the second letter, so this must mean that U stands for the light shading of the pentagon. The solid black shading is therefore S. So the final answer is KS.)

Missing Sequence

1. **B** - (As you move from left to right two changes happen each time; the triangles alternate between left and right and the squares alternate between removing one line from the bottom and one line from the top. In the missing square one line should be taken from the top compared to the previous square and the top triangle should be on the left, so the answer is B.)
2. **B** - (The squares keep alternating with two changes each time. The chevron is reflected through a horizontal line through the centre of the square and the circle changes colour from dark to white or white to dark. So, the answer must be B.)

Missing Square

1. **B** - (From the top row you can see the only difference is that the shape from the left square has got smaller. By applying this to the bottom row you can work out that the missing square is B.)
2. **A** - (The shapes in the bottom row have been reflected through a horizontal line. If you reflect the middle square in the top row through the same horizontal line then you will find that the answer is A.)

Most Like

1. **B** - (Both of the squares on the left are symmetrical through a line vertically down the centre of the squares. The only answer that is symmetrical through a line vertically down the centre of the square is B.)
2. **D** - (The similarities between the two shapes on the left are that there are two vertical lines inside and a circle on the outside of the shape. The only shape with two vertical lines is D, so the circle is actually irrelevant. So the answer is D.)

Nets

1. **A** - (You can see from the cube that the top of the pentagon should line up with the white square being above. Looking at the lines as vertical rather than horizontal, the left hand side of them should line up with the white square. From this information, you can see that the only possible answer is A.)

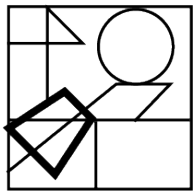
2. **A** - (From the net you can see that the square with the single circle in has the black square below it and also has the lightly shaded square to its right. From this information, you can see that the answer is A.)

3D Shapes

1. **C** - (C must be the correct answer because all the other answer options feature a gap that is clearly not there. The square in the top right corner of the 2D view can be deduced to be there even though it cannot be seen from the viewpoint of the 3D shape.)
2. **A** - (The shape must be made from two 'T' shaped blocks and two cuboids, which lie horizontally to create a structure that resembles stairs.)

Hidden Shape

1. **C** -



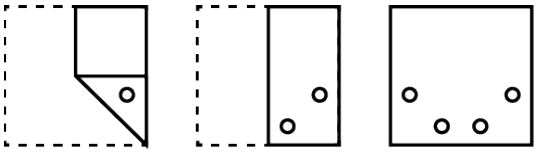
2. **B** -

Fold Along the Dotted Line

1. **A** - (The right hand portion of the shape is folded behind the left hand portion, with the effect that the right hand portion is no longer visible.)

Hole Punch

1. **D** -



Shaded Nets

1. ***C*** - (You can see from the net that the shape must have a white square base. C is the only option with a white square base.)